

- **Group user** – is one member of a group, all the members of which get the same permissions for some particular content.
- **Other user** – is the one who is not an owner of content. Write and Execute permissions are not given to this user, unless really required.

A **group** is a collection of users to whom common permissions are allocated for any content. There is only one user owner for any content but, when we need to allow multiple users to access or modify the content by working as shared owners of that content, group ownership comes into the picture.

There are two types of groups:

- Primary group is the default group of a user, which is mandatory and grants group ownership. It is mapped in `/etc/passwd` file with the corresponding user.
- The secondary group allows multiple users to become members in order to provide them content-accessing permissions. Members of this group do not get ownership of content.



Tips: 1. Adding and managing users seems to be very simple in graphical mode, but for administrators it is recommended to use commands on the terminal.

2. One user can have only one primary group, which is called his private group. However, users can enjoy secondary membership of multiple groups. This means that a user can be a member of multiple groups but only through secondary group membership.

Creating users and groups with default configurations

Let us add two new users named `u1` and `u2` with default configurations and issue a password to `u1` only.

```
#useradd <username>
#passwd <username>
```

```
For Example:
#useradd u1
#passwd u1
#useradd u2
```

By default, a new group is created with the same name as the username and a `/bin/bash` shell is assigned. Also, the user gets a directory with this name inside `/home` and receives some predefined files from the `/etc/skel` directory for starting work.

A new group can be added as follows:

```
#groupadd <groupname>
```

```
For example:
#groupadd mygroup
```

Information and configuration parameters of users and groups

Information about existing users is present in `/etc/passwd` and has the following format:

```
<username>:<auth_check>:<userid>:<groupid>:<Comments>:<Homedir>:<Shell>
```

The various fields separated by ':' in this file are:

- **username** is a unique name issued to the person who is to use the Linux interface.
- The **auth_check** field only denotes whether this user has a password or not. The actual password is stored in `/etc/shadow` file.
- The **userid** is a unique number issued to identify a user. 1 - 499 are reserved for system applications and 499 onwards are issued to new users managed by the administrator. With default configurations, this ID is issued in an incremental manner.
- The **groupid** is a unique number issued to identify a group of users. 1 - 499 are reserved for system applications and 499 onwards are issued to new groups managed by the administrator. With default configurations, this ID is issued in an incremental manner through the `/etc/group` file.
- The **Comments** field is optional and contains information about the user.
- The **Home directory** is the default path where users can manage their documents and keep them safe from other users.
- **Shell** is an environment provided as an interface to the user. **Shell** decides which tools, utilities and facilities are to be provided. Examples are `ksh`, `bash`, `nologin`, etc.

Password related information of the user is automatically managed in the `/etc/shadow` file. Here, passwords are saved in encrypted form.



Tips: Log in as the root user (the superuser with unlimited privileges) to access and manage these files.

Let us have a look at the `/etc/passwd` file by using the `tail` command.

You can see that system user '`tcpdump`' has IDs less than 500 while `u1` and `u2` have IDs that are greater than 500.

Let us have a look at the `/etc/shadow` file by using the `tail` command.

Here, `u1` has an encrypted password while `tcpdump` and `u2` have null passwords.

```
[root@localhost Desktop]# tail -4 /etc/shadow
sshd:!:16388:!!!!:
tcpdump:!:16388:!!!!:
u1:$1$j0qxgZmKSEd1KHytAGH7awmQVUR5Cg.:16388:0:99999:7:::
u2:!:16388:0:99999:7:::
[root@localhost Desktop]#
```

Figure 1: Output of the tail command